

German University in Cairo

Media Engineering and Technology (MET) Department

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DMET 901 – Computer Vision

Milestone 1 Guidelines

1. Purpose

In this project you will identify a real computer vision problem, select a suitable dataset, and implement a working solution for it.

Milestone 1 establishes the foundation: **what problem you are solving, what data you will solve it on, and what model you will start from**. Every later milestone builds on this, so the choices made here must be defensible.

3. What to Present in Milestone 1

3.1 Problem statement and motivation

State the vision task precisely, not in general terms. Your slides must identify:

- **Task type:** classification, object detection, semantic or instance segmentation, tracking, pose estimation, OCR, image retrieval, etc.
- **Input:** RGB image, a video frame sequence, multi-spectral, etc.
- **Output:** a class label, a set of bounding boxes with classes, a per-pixel mask, etc.
- **Motivation:** why this problem matters, who would use the system, and what a human currently does instead.

3.2 The dataset

This section should take the largest share of your presentation. Cover each of the following:

- **Source and licensing:** where the dataset come from
- **Size and composition:** number of images, number of classes, images per class, image resolutions, file formats.
- **Annotations:** what the ground truth actually contains, the annotation format.
- **Show the data.** Display real sample images from your dataset: at least one per class. A class-distribution plot is expected.
- **Dataset challenges:** class imbalance, small objects, occlusion, motion blur, lighting and background variation, domain shift between training and deployment conditions, label noise, duplicate images.
- **Justification:** why this dataset is an appropriate proxy for the problem in Section 3.1, and where it is not.

3.3 Model and tooling

- **Architecture:** the model you will start from and why it fits the task and the size of your data (a CNN backbone, a YOLO variant, U-Net, a ViT, or a classical pipeline).

- **Training strategy:** train from scratch, fine-tune pretrained weights, or use a frozen backbone with a new head — and what the size of your dataset implies about that choice.
- **Libraries and tools:** PyTorch or TensorFlow, OpenCV, Ultralytics, Albumentations, timm, Kaggle or Colab, together with your hardware constraints.

3.4 Evaluation plan

State the metrics you will report and why they suit your task and class distribution — for example accuracy versus macro-F1 under imbalance, mAP@0.5 and mAP@0.5:0.95 for detection, IoU or Dice for segmentation

4. Format

- 5–7 minutes per team. One team member presenting is sufficient.
- Slides are submitted on the day of the presentation.
- No results are required at this stage. A loaded dataset, a printed class distribution, and sample images displayed from your own code are enough to show that the data is real and usable.

5. How This Milestone Is Graded

You are assessed on the clarity and specificity of the problem definition and on **how well you understand your data** — not on model performance, which is not expected yet.

A team that has honestly identified serious problems in its dataset scores better than a team that reports none. Vague statements such as "we will detect objects in images" or "the dataset is from Kaggle" will lose marks; statements tied to evidence you can show on a slide will not.

6. Timeline

The presentations will take place from September 26th to October 19th. Your exact presentation date and time will be posted on CMS